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B.Tech/M.Tech (Integrated) DEGREE EXAMINATION, JULY 2025

Sixth Semester

21ECC302T - ANALOG AND DIGITAL COMMUNICATION

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40^{th} minute.
ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

	Question	Marks	BL	CO
1.	The modulation index in AM is given by (A) Ac/Am (B) Am/Ac (C) Ac× Am (D) Am/2Ac	1	2	1
2.	A 1 MHz carrier is amplitude modulated by a 5 kHz message signal with a modulation index of 0.5. Calculate the total bandwidth of the AM signal? (A) 5 kHz (B) 15 kHz (C) 10 kHz (D) 20 kHz	1	3	1
3.	Select the essential components in an envelope detector (A) Mixer and oscillator (B) Diode and Capacitor (C) Transistor and Inductor (D) Frequency multiplier and filter	1	2	1
4.	In frequency modulation (FM), increasing the modulation index leads to (A) Decrease in bandwidth (B) Increase in bandwidth (C) No effect on bandwidth (D) Reduction in carrier frequency	1	2	1
5.	In a superheterodyne receiver, the function of the local oscillator is to: (A) Amplify the received signal (B) Convert the received signal to an intermediate frequency (C) Remove unwanted noise (D) Detect the message signal	1	1	2
6.	Identity the use of amplitude limiter in FM receivers (A) Demodulation (B) Amplification (C) Remove amplitude variations due to noise (D) Filtration	1	1	2
7.	The purpose of pre-emphasis in FM transmission is to (A) Reduce noise at higher frequencies (B) Improve low-frequency response (C) Increase modulation index (D) Reduce transmission bandwidth	1	2	2
8.	A superheterodyne receiver receives a signal at 1 MHz. If the intermediate frequency (IF) used is 455 kHz, what should be the local oscillator frequency? (A) 545 kHz (B) 1.455 MHz (C) 455 kHz (D) 1.455 MHz or 545 kHz	1	3	2
9.	A PCM system uses an 8-bit quantizer and a sampling rate of 10 kHz. What is the total bit rate of the system? (A) 160 kbps (B) 8 kbps (C) 10 kbps (D) 80 kbps	1	3	3

10. The purpose of an Eye Pattern in digital communication is to 1 2 3
(A) Analyze and visualize inter-symbol interference (ISI)
(B) Improve the frequency response of a system
(C) Reduce the signal-to-noise ratio
(D) Increase transmission bandwidth

11. In a Matched Filter Receiver, the main purpose of the matched filter is to 1 2 3
(A) Minimize noise power in the received signal
(B) Maximize the signal-to-noise ratio (SNR) at the sampling instant
(C) Reduce bandwidth usage
(D) Eliminate phase distortion

12. The number of voltage levels used to represent a PWM signal is 1 2 3
(A) 1
(B) 3
(C) 0
(D) 2

13. In Binary Phase Shift Keying (BPSK), the phase of the carrier signal is shifted by 1 2 4
(A) 45°
(B) 90°
(C) 180°
(D) 360°

14. In a BFSK system, the two carrier frequencies used are 100 kHz and 120 kHz. What is the frequency deviation? 1 3 4
(A) 10 kHz
(B) 20 kHz
(C) 30 kHz
(D) 40 kHz

15. Which passband modulation technique uses two different frequencies to represent binary data? 1 2 4
(A) BPSK
(B) QPSK
(C) BFSK
(D) QAM

16. In QPSK , the phase difference between the different constellation waveforms is 1 2 4
(A) $\pi/2$
(B) $2\pi/3$
(C) π
(D) $\pi/4$

17. The unit of information in Shannon's theory is 1 2 5
(A) Joule
(B) Bit
(C) Hertz
(D) Decibel

18. According to Shannon's Channel Capacity Theorem, channel capacity C is given by 1 3 5
(A) $C = B * \log_2(1 + SNR)$
(B) $C=B/SNR$
(C) $C=SNR\times B$
(D) $C= SNR/B$

19. Entropy in information theory measures 1 2 5
(A) The noise in a signal
(B) The randomness or uncertainty in a message
(C) The power of the transmitted signal
(D) The number of errors in a transmission

20. Huffman coding is used for 1 2 5
(A) Signal amplification
(B) Increasing bandwidth
(C) Data compression
(D) Error detection

PART - B (5 × 8 = 40 Marks)

Answer **all** Questions

Marks BL CO

21. (a) (i) Justify the need for modulation in communication systems with suitable examples. 8 2 1
(ii) Illustrate about Amplitude Modulation (AM) with appropriate diagrams.

(OR)

(b) (i) Explain the working of a Varactor Diode Modulator for generating Narrowband FM.
(ii) Describe the Envelope Detector method for AM demodulation with a neat diagram.

		Marks	BL	CO
22.	(a) (i) Compare and contrast Low-Level and High-Level AM transmitters. (ii) Illustrate the working of a Direct FM Transmitter with a block diagram. **(OR)** (b) (i) Describe the working principle of Superheterodyne Receiver. (ii) Justify the need for Pre-emphasis and De-emphasis circuits in FM communication.	8	2	2
23.	(a) (i) Discuss how an Eye Pattern helps in detecting Inter-Symbol Interference. (ii) Justify the importance of a Matched Filter Receiver in minimizing probability of error. **(OR)** (b) i) Differentiate between PAM, PWM, and PPM with suitable waveforms. ii) A PCM system is designed to digitize an analog signal with a maximum frequency of 4 kHz. The system uses a 12-bit quantizer. (1) Determine the minimum sampling rate required according to the Nyquist theorem. (2) Calculate the total bit rate of the system.	8	3	3
24.	(a) (i) Explain the Passband Transmission System Model with a neat block diagram. (ii) Justify the importance of Quadrature Amplitude Modulation (QAM) in modern digital communication. **(OR)** (b) (i) Illustrate the working of QPSK modulation with a signal space diagram. (ii) A QPSK system operates over an AWGN channel with a bit rate of 20 kbps. The system transmits with a total power of 40 mW and has a channel bandwidth of 10 kHz. The noise power spectral density is $No/2=5\times10^{-10}$ W/Hz. (1) Determine the energy per bit. (2) Calculate the SNR per bit in dB.	8	2	4
25.	(a) (i) Define entropy and explain its significance in information theory. (ii) A source transmits four symbols S1,S2,S3,S4 with probabilities 0.4, 0.3, 0.2, and 0.1 respectively. Calculate the entropy of the source. **(OR)** (b) Consider a binary symmetric channel with an error probability of 0.1. If the mutual information between input and output is given by $(X;Y)=H(X)-H(X\|Y)$ Calculate the mutual information assuming a uniform input distribution.	8	3	5

PART - C (1 × 15 = 15 Marks)

Answer **any 1** Questions

		Marks	BL	CO
26.	(i)Derive the expression for Narrowband Frequency Modulation and explain how it differs from Wideband FM in terms of bandwidth and applications. (ii) Explain the working of the Foster-Seely discriminator for FM demodulation with a neat diagram	15	3	2
27.	i) Explain Shannon's Channel Capacity Theorem and derive the formula for channel capacity. (ii) A discrete memory less source has 5 symbols x1,x2,x3,x4,x5 with probabilities 0.4, 0.19, 0.16, 0.15 and 0.1. Find the Huffman code for the source.	15	3	5

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